

L9b: American Options, CRR Pricing, and the Value of Early Exercise

CHEME 5660 Cornell University

Copyright Jeffrey Varner 2026. All Rights Reserved

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

L9a priced contracts exercisable only at expiration. Today the holder gets a **choice of when to exercise**. We keep the same terminal payoffs, price both exercise styles on one lattice, and measure what that extra right is worth.

Today's concepts

- **Contrast European and American exercise**: explain why identical terminal payoffs can carry different premiums when one holder may exercise early.
- **Price both styles on one CRR lattice**: apply the European and American backward-induction rules under identical inputs and verify that the American value is never smaller.
- **Explain the premium difference**: separate intrinsic value, extrinsic value, and the early-exercise premium, and identify when the American premium is strictly higher.

Lectures and Examples

Lecture: the American exercise set, the CRR lattice, the two backward-induction rules, the three value differences, and the conditions under which American exercise is actually worth something.

Example 1: the two-step textbook put. Price both styles on one lattice and inspect the node where exercise beats continuation.

Example 2: an AMD option chain. Price the whole chain with CRR under both rules and compare with quoted bid-ask intervals.

Example 3: an NVDA call. Split the premium into intrinsic and extrinsic value across strike and time to expiration.

Same Terminal Payoff, Different Exercise Set

For strike $K > 0$ and underlying price $S \geq 0$, with $x^+ := \max(x, 0)$, the immediate-exercise values are:

$$h_c(S) = (S - K)^+, \quad h_p(S) = (K - S)^+.$$

- **European**: receives this payoff only at expiration T .
- **American**: may choose any exercise time $\tau \leq T$, which includes the strategy of never exercising early.

The American strategy set **contains** the European one, so under identical model inputs:

$$V_0^{\text{AM}} \geq V_0^{\text{EU}}.$$

The inequality need not be strict.

Build the Cox–Ross–Rubinstein Lattice

Split maturity T into N steps of length $\Delta t = T/N$. With continuously compounded risk-free rate g_y and annualized volatility $\sigma > 0$, CRR sets the move sizes and the pricing weight q :

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad \boxed{q = \frac{e^{g_y\Delta t} - d}{u - d}}.$$

- This is **L3b's one-step result** for a continuous rate: there $R_f = e^{g_y\Delta t}$ and $q = (R_f - d)/(u - d)$.
- q is a **pricing weight**, not a forecast. L3b keeps p for the real-world up probability; only q appears today.
- No arbitrage requires $d < e^{g_y\Delta t} < u$, equivalently $0 < q < 1$. If that fails, **refine the step**, never clip q .
- With a continuous dividend yield δ , replace $e^{g_y\Delta t}$ in the numerator by $e^{(g_y - \delta)\Delta t}$.

One Lattice, Two Backward-Induction Rules

Index a node by its level $t \in \{0, \dots, N\}$ and its number of up moves $k \in \{0, \dots, t\}$, so $S_{t,k} = S_0 u^k d^{t-k}$, as in L3b and L4a. At expiration both styles agree, $V_{N,k} = h(S_{N,k})$, where h is whichever of h_c , h_p applies. For $t < N$ the discounted continuation value is:

$$C_{t,k} = e^{-g_y \Delta t} \underbrace{\left[qV_{t+1,k+1} + (1-q)V_{t+1,k} \right]}_{\text{pricing expectation}}.$$

The exercise convention changes **exactly one line**:

$$\boxed{V_{t,k}^{\text{EU}} = C_{t,k}^{\text{EU}}} \quad \boxed{V_{t,k}^{\text{AM}} = \max\{h(S_{t,k}), C_{t,k}^{\text{AM}}\}}.$$

Exercise where $h > C$, wait where $C > h$. Equality is indifference: the premium does not depend on the tie-break, the reported decision does.

A Two-Step Put Makes the Difference Visible

Take $S_0 = 50$, $K = 52$, two one-year steps, $u = 1.2$, $d = 0.8$, and $g_y = 5\%$, so $q \approx 0.6282$. Since $ud = 0.96 \neq 1$ this is a **generic** binomial tree, not a CRR calibration; the two recursions are unchanged either way.

At the one-year down node $S = 40$, exercising pays 12 while waiting is worth:

$$e^{-0.05}[0.6282(4) + 0.3718(20)] \approx 9.46.$$

The American holder **exercises**; the European holder must wait. Rolling both trees to the root:

$$V_0^{\text{EU}} \approx 4.19, \quad V_0^{\text{AM}} \approx 5.09.$$

The early-exercise right is worth about 0.90 USD/share here.

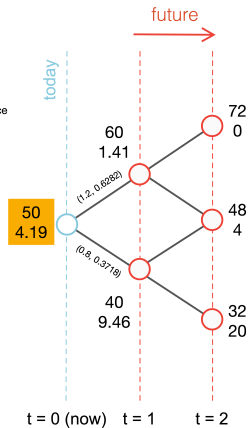
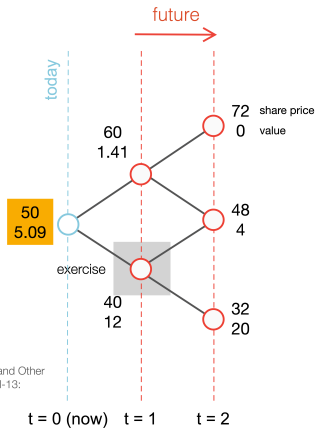
Example 1: `CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb`

One Node Carries the Whole Difference

Example: Hull 13.4 and 13.5

American Put contract

European Put contract



Reference: John C. Hull "Options, Futures, and Other Derivatives" 9th edition, 2015. Pearson, ISBN-13: 978-0133456318

CornellEngineering

Only the down node differs: American 12, European 9.46.

Three Value Differences Answer Three Questions

Let V be the optimally managed value, h intrinsic value, and C continuation value. At an interior node, that is for $t < N$:

$$\underbrace{X_{t,k} = V_{t,k} - h(S_{t,k})}_{\text{extrinsic value}} = (C_{t,k} - h(S_{t,k}))^+ \geq 0,$$

because $\max\{h, C\} - h = \max\{0, C - h\}$. Continuation is undefined at expiration, where $V_{N,k} = h(S_{N,k})$ gives $X_{N,k} = 0$ directly. Across styles:

$$\underbrace{A = V^{\text{AM}} - V^{\text{EU}}}_{\text{early-exercise premium}} \geq 0.$$

- C : what is waiting worth?
- X : how far does the managed contract exceed immediate exercise?
- A : what does American flexibility add over the same European contract?

When Is the American Premium Strictly Higher?

- **Non-dividend calls, $g_y \geq 0$:** early exercise is never strictly better, so $V_c^{\text{AM}} = V_c^{\text{EU}}$. Paying K early and discarding optionality cannot help. For $g_y > 0$ waiting is strictly better; at $g_y = 0$ the holder can be indifferent.
- **Puts:** early exercise can be optimal. Receiving K sooner has financing value, but that has to **exceed the optionality given up**, which is why the put must also be sufficiently in the money.
- **Dividend-paying calls:** exercising before an ex-dividend date can pay when the dividend beats retaining the option and deferring K .
- **Negative rates, discrete dividends, funding spreads, constraints:** each needs its own model before any exercise rule follows.

American style guarantees only the **weak** inequality.

Examples 2 and 3: AMD chain premiums; NVDA intrinsic and extrinsic value

Optional Advanced Material

Advanced: [advanced/crr_factors/](#)

CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

Optional, not a prerequisite for L10a. Today's lecture **quotes** $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$; the notebook derives them. It writes the one-step log return as a two-point variable and matches its variance to the diffusion variance $\sigma^2\Delta t$, and it is explicit about the price: $ud = 1$ is a **normalization the modeler chooses**, not something the diffusion forces, the equal weights used to size the jump are **not** the pricing probability q , and the variance match holds only to leading order, discarding a term of order Δt^2 . It closes with the domain $|g_y|\sqrt{\Delta t} < \sigma$ that keeps $0 < q < 1$.

Notebook: [lectures/week-9/L9b/advanced](#), also linked from the lecture notebook.

Summary

Today we took L9a's European price as a benchmark, added the American exercise decision on the same lattice, and measured what that extra right is worth.

Key takeaways

- **American style expands the exercise set:** the American holder can always choose to wait until expiration, so the American premium is never smaller than the European premium under identical inputs.
- **Backward induction creates the difference:** the European recursion always continues, the American recursion takes the larger of intrinsic and continuation value, and that one line is the whole distinction.
- **Higher is not automatic:** on the AMD chain the American and European call premiums agree exactly, while the puts carry a positive early-exercise premium that reaches about 2.30 USD/share at the deepest in-the-money strike.

Next, L10a asks how a premium responds locally to spot, volatility, time, and interest rates through the Greeks.